

Remapping Time in Augmented Reality

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Abstract

Video see-through augmented reality (VST AR) mediates the physical world through a camera capture–processing–display pipeline, making visual time computationally controllable. Yet this capability as a design consideration remains underexplored. We define and formalize *time remapping* as the intentional, real-time manipulation of how visual world time is presented to the user, and model it as a first-order dynamical system that balances delay accumulation and recovery. To evaluate its perceptual effectiveness, we conducted a within-subjects study ($N = 21$) in which participants identified briefly visible playing cards under baseline and remapped conditions, varying temporal anchor (self vs. external) and control (user vs. system). Time remapping approximately doubled identification accuracy without significantly reducing comfort, with gains consistent across all remapped conditions. Building on these findings and qualitative interviews, we derive a six-dimensional design space and instantiate it through three application prototypes. This paper establishes time remapping as a practical design material, introducing programmable time as a new dimension for AR experience design.

CCS Concepts

• **Human-centered computing** → **Mixed / augmented reality**; *Interaction design theory, concepts and paradigms.*

Keywords

Augmented Reality, Video see-through, Time remapping, Temporal manipulation, Computational interaction

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1 Introduction

Augmented reality (AR) is often framed as a medium for making space programmable. In video see-through (VST) AR, this programmability extends beyond space: because the world is captured, processed, and re-displayed through a camera pipeline, the timing of visual experience also becomes computationally controllable.

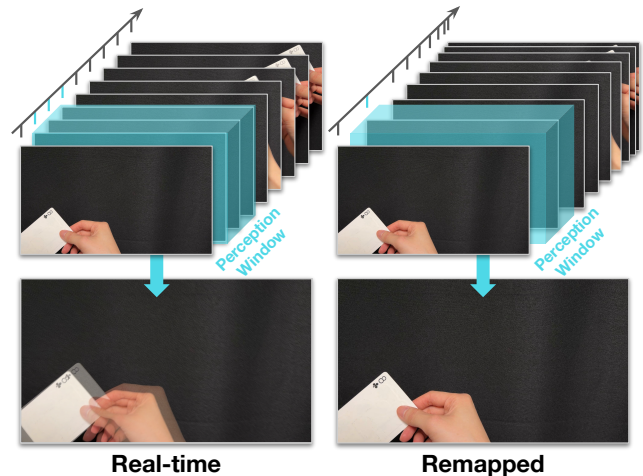


Figure 1: Video see-through AR presents the real world as a sequence of frames over time. In real-time viewing (left), frames are evenly spaced, and multiple frames may fall within a short temporal window during fast motion. Time remapping (right) alters the temporal spacing of frames and can reduce the number of frames within such a window. The bottom images illustrate the resulting perceived outputs.

Just as this mediation enables spatial manipulations such as imperceptibly switching environments [77], producing virtual-physical decoupling illusions [37], substituting objects in real time [38], or removing elements via diminished reality [17], it also makes visual time manipulable.

Yet this capability as a design consideration remains underexplored. Real-time presentation is often taken as the default, but it is not always the most meaningful temporal presentation for perception or interaction. Departing from real time can reveal fleeting details (Fig. 1), alter how motion is understood, amplify the felt impact of events, or change how objects and actions are experienced, which are effects already common in slow motion, replay, and other temporal shaping outside interactive systems. This suggests a broader opportunity for VST AR: once visual time becomes programmable, how should the flow of displayed time be designed?

We propose *time remapping* as a design material for VST AR: continuous, real-time, frame-level control over how captured world time is mapped onto displayed time, formalized as a first-order dynamical system that provides a simple and interpretable model for introducing, regulating, and recovering temporal offset.

This raises three related questions: (RQ1) **can time remapping be effectively operationalized**, (RQ2) **how does it impact**



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user perception and interaction, and (RQ3) what are the main design dimensions for time remapping across diverse application scenarios?

To answer these questions, we conduct a within-subjects VST AR study ($N = 21$) in which participants identified playing cards that are only visible during fast motion under baseline and remapped conditions. We probe two initial design dimensions: temporal anchor (whether remapping is coupled to self-generated or externally generated motion) and control (user- vs. system-controlled).

Our results show that time remapping can approximately double card identification accuracy without significantly reducing comfort, with gains robust across both control strategies. Qualitative findings from post-study interviews reveal a broader design space comprising six dimensions: *Anchor*, *Control*, *Parametrization*, *Scope*, *Representation*, and *Transparency*, capturing how remapping is triggered, implemented, and communicated. We further instantiate this design space through three demonstration applications: hybrid-control speed regulation for physical training, depth-dependent slowdown for pseudo-textured water, and forward-sampled speed exaggeration for high-speed slashing.

This paper makes three contributions:

- We introduce time remapping as a programmable design material in VST AR and formalize it as a real-time dynamical system with a first-order linear ordinary differential equation (ODE).
- We provide empirical evidence that time remapping improves dynamic acuity without compromising comfort, across different anchors and control strategies.
- We derive a multi-dimensional design space from qualitative findings, characterizing how time remapping can be designed and experienced in AR.

2 Related Work

2.1 Temporal Manipulation in AR/VR

Virtual reality (VR) and video see-through AR afford direct manipulation of time because the entire scene is rendered computationally. Prior work has explored temporal experiences such as revisiting recorded events in immersive environments [26], visualizing summaries of temporal changes in mixed reality [18], and rendering immersive event playback under causality constraints [24]. Bringing similar capabilities to optical see-through AR systems is more challenging, as users view the physical world directly. Reconstruction-based approaches such as Remixed Reality [46] address this by enabling users to pause and revisit captured scenes through live 3D reconstruction. Other work has used screen-based mediated reality devices (e.g., MIRAGE) to manipulate the temporal relationship between users' actions and their visual embodiment, examining consequences for agency and body ownership [61].

A small number of prior systems have also introduced deliberate temporal departure from real time in AR. Knierim et al. [41] presented an early proof-of-concept of AR slow-motion using frame repetition in a displayed video stream on the optical see-through HoloLens. Muramoto et al. [50] achieved slow-motion without real-time deviation in a custom VST system by exploiting binocular integration to present differentially timed streams to each eye. Different from these prior explorations, which realize slow motion

through fixed, single-purpose mechanisms, we treat time remapping as a general, parameterized framework: instead of one specific effect, it exposes continuous control over how captured time maps onto displayed time, from which such fixed behaviors emerge as particular configurations. We formalize this as a dynamical system and characterize the resulting range of designs through an empirical study.

A parallel line of research has examined timing in AR/VR through the lens of system performance and temporal fidelity [27, 32, 59, 65, 73]. A central concern is rendering latency: techniques such as camera time warp reproject previously rendered frames based on the current head pose to compensate for motion-to-photon delay [20, 25, 40, 51, 71], motivated largely by the demonstrated link between latency and cybersickness [15, 57, 66, 67, 70]. However, existing studies have mostly examined fixed static delays or jitter patterns that mimic rendering latency in practice, leaving open whether structured and intentional temporal manipulation produces similar discomfort effects, which is a gap our comfort and motion sickness measures directly address.

2.2 Dynamic Acuity and Perceptual Augmentation

Dynamic visual acuity (DVA) is the ability to resolve fine spatial detail in moving or briefly visible targets [9, 14, 16, 58]. Unlike static acuity, DVA degrades sharply with target velocity and exposure duration as retinal image stabilization becomes insufficient [9, 75]. A card-dealing task adopted in the present study operationalizes this limit directly: brief, motion-dependent target visibility creates a realistic condition where real-time presentation is perceptually insufficient and temporal augmentation can produce measurable gains.

VST AR affords a particularly flexible perceptual intervention because the physical world is re-presented through a camera pipeline, allowing the system to alter visual presentation at the pixel and frame levels. Prior VST systems have exploited this to reconfigure environments [77], substitute objects [37], and suppress scene elements [17]. Similarly, time remapping makes use of the flexibility of VST AR to control the temporal dimension of visual presentation.

Additionally, altering temporal presentation has also been shown to influence perception and behavior in VR and video-based systems [52, 61, 72, 74], with documented psychophysical consequences including effects on perceived causality, agency, and motor timing [6, 13, 31]. A related thread uses visual dominance over proprioception under visuomotor conflict to design perceptual effects in VR [62], underlying spatial manipulations such as haptic retargeting [3], redirected touching [42], and pseudo-haptic feedback [45]. Our observations suggest this dominance extends to AR: when the visual presentation of one's own motion is modified, motor output may tend to adjust accordingly.

2.3 Time as an Interaction Resource

Beyond AR/VR, HCI research has long treated time not only as something to measure but as something to design. Slow-motion capture and replay have been used for sports learning and movement analysis [33]; adaptive playback systems modulate temporal flow to support physical activity and reduce cognitive effort [19, 68].

Time-warped visualization has also been applied in robotics to support trajectory understanding and interpretability [69]. Together, these works establish temporal shaping as a productive interaction resource: one that can reveal, emphasize, or reorganize events in ways that support user understanding.

A complementary consideration is how different temporal shaping may affect users' own perception of time. Human time perception is highly malleable [43], influenced by factors such as emotion [21–23, 28, 54], motion [2, 11, 29, 44, 56], attention [10, 12, 80], and virtual environment immersion [30, 48, 53, 63]. Since our tasks involve temporally manipulated motion, time perception may be incidentally affected. We therefore include a duration reproduction task [5] as an exploratory measure.

3 Time Remapping

In video see-through (VST) AR, the world is viewed through a camera–processing–rendering pipeline rather than directly. If we store recently captured frames in a buffer, the system can choose which captured moment to display at each instant. We refer to this intentional, frame-level control over how displayed visual time is mapped to physical time as **time remapping**.

3.1 Formalization

Let $\{(I_i, \tau_i)\}_{i=1}^N$ denote the N most recent buffered frames, where I_i is the image and τ_i its capture timestamp. At physical time t , the system specifies a *source time* t_s and displays the buffered frame whose timestamp is closest:

$$t_s = g(t), \quad i^*(t) = \arg \min_i |\tau_i - t_s|. \quad (1)$$

Here, $g(t)$ is the **time-mapping function**, defining which moment of the captured world is shown at time t . This allows displayed time to slow, pause, or accelerate relative to physical time.

Playback rate. The instantaneous rate of displayed time is

$$r(t) = \frac{dg(t)}{dt}. \quad (2)$$

When $r(t) = 1$, the display follows real time; when $r(t) < 1$, visual time slows; when $r(t) = 0$, visual time pauses; and when $r(t) > 1$, visual time advances faster to reduce previously accumulated delay.

Temporal offset. The difference between physical and displayed time is

$$d(t) = t - g(t). \quad (3)$$

Together, $g(t)$, $r(t)$, and $d(t)$ describe selected world time, playback speed, and accumulated offset. Increasing $d(t)$ makes the display show an older buffered moment, while reducing $d(t)$ returns the display toward the present. When $d(t) < 0$, the display shows a future moment, realizable through forward sampling as discussed in Sec. 6.

3.2 Remapping Dynamics

We model the temporal offset $d(t)$ as a dynamical system governed by a first-order linear ODE for three reasons: (1) it is the minimal model capturing the two relevant directions of temporal offset (accumulation and recovery); (2) recovery proportional to accumulated offset $\alpha(t)d(t)$ ensures smooth resynchronization without explicit

smoothing logic; and (3) the single state variable $d(t)$ integrates trivially in real time. Higher-order models would introduce dynamics such as overshoot that have no meaningful analog in temporal offset.

Formally:

$$\dot{d}(t) = u(t) - \alpha(t)d(t), \quad (4)$$

where $u(t) \geq 0$ is the **offset injection rate** and $\alpha(t) \geq 0$ is the **recovery gain**. Intuitively, $u(t)$ accumulates delay and $\alpha(t)$ reduces it. Because $-\alpha(t)d(t)$ pulls the offset toward zero from either sign, the same system handles past ($d(t) > 0$), present ($d(t) = 0$), and future ($d(t) < 0$) symmetrically. Intuitively, the system behaves like a spring that does not oscillate: u pushes the display away from the present while α pulls it back, settling at real time without overshooting into a false future.

Combining Eq. 2, Eq. 3 and Eq. 4, the playback rate becomes

$$r(t) = \dot{g}(t) = \frac{d}{dt}(t - d(t)) = 1 - \dot{d}(t) = 1 - u(t) + \alpha(t)d(t). \quad (5)$$

Thus, increasing $u(t)$ slows playback below real time, whereas positive $\alpha(t)$ drives the display back toward or above real time by recovering accumulated offset.

3.3 Initial Design Dimensions

Time remapping introduces many possible design choices. As a first empirical probe of this space, we focus on two dimensions motivated by prior work on embodied interaction and automation: *temporal anchor* and *control*. These capture what the remapping is centered on and who regulates it.

Anchor. We borrow the notion of anchor from spatial AR design, where content is classified by its reference frame as body-, head-, or world-anchored [1, 49, 60], extending this concept from *where* augmentation is situated to *what* the temporal manipulation is centered on.

Time remapping may be centered on different elements of an interaction. Prior work suggests an important distinction between self-generated motion, which is coupled to sensorimotor prediction and efference copy, and externally observed motion, which is not [64, 73, 79]. We therefore consider two anchors: *body-centered* remapping, coupled to the user's own actions, and *world-centered* remapping, applied to externally generated events. These anchors represent fundamentally different interaction contexts: body-centered remapping introduces a visual-motor mismatch that world-centered remapping does not, making them qualitatively distinct as design targets.

Control. A second dimension concerns who initiates or regulates remapping. Prior work has contrasted user control, which improves predictability and perceived agency, with system automation, which reduces effort and enables automatic response to events [19, 36, 68]. We consider *user-controlled* remapping, in which users explicitly trigger slowdown, and *system-controlled* remapping, in which the system automatically detects relevant motion events and applies remapping.

4 Evaluation Study

We conducted a within-subjects study to evaluate whether time remapping improves dynamic acuity while remaining tolerable

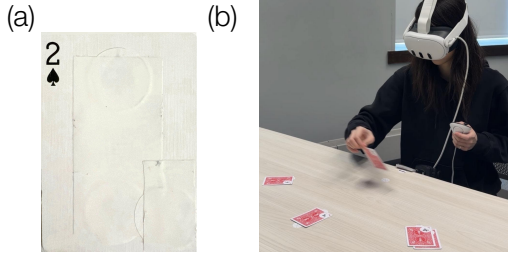


Figure 2: (a) An example of the cards used in the study, only the suit and rank at one corner is revealed. (b) The study setup in the user-controlled body-centered task. The participant deals cards to 3 marked target locations while using the controller to trigger the remap with the other hand.

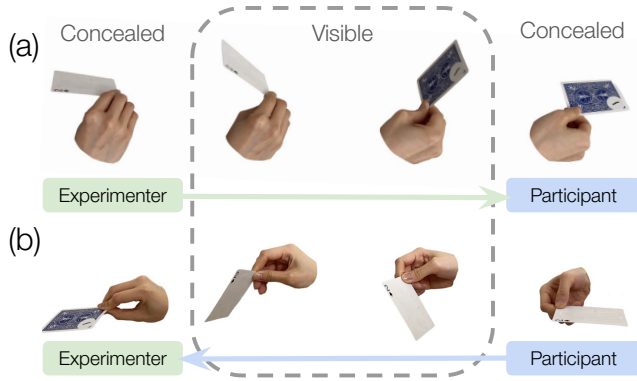


Figure 3: The hand motion of card dealing in the (a) world-centered task and (b) body-centered task. The cards are concealed under a deck at the start and are face down in the end, only visible during the dealing motion.

in VST AR, and to probe two initial design dimensions of time remapping: temporal anchor and control.

4.1 Study Design

We designed two card-identification tasks corresponding to two temporal anchor contexts: *world-centered* and *body-centered*.

World-centered task. Participants observed an experimenter dealing cards to three marked target locations. Each card began concealed (face-up at the bottom of the deck); its face became briefly visible during the dealing motion, and it landed face-down at the target (Fig. 3a). Participants reported the card identity after each deal.

Body-centered task. Participants dealt cards themselves to three different target locations, under otherwise identical conditions: each card concealed at the start, briefly visible during the motion, and landed face-down (Fig. 3b).

In both tasks, deals rotated across the three target locations (left, center, right) to reset visual attention across trials.

Each task included three conditions: **Baseline**: no time remapping ($r(t) = 1$); **System-controlled**: remapping triggered automatically by the system; **User-controlled**: remapping triggered by the participant.

This yielded a $2 (\text{Task}) \times 3 (\text{Condition})$ within-subjects design. Condition order was counterbalanced independently for each task using a balanced Latin square (6 orders for 3 conditions), with participants assigned to orders as evenly as possible across the sample ($N = 21$). The world-centered task was always experienced first to allow participants to familiarize themselves with the remapping mechanism before the more demanding body-centered task. Cross-task subjective comparisons should be interpreted with this ordering in mind. Since no cross-task comparisons are made in our performance analyses, this learning effect from fixed ordering does not introduce a confounding factor.

4.2 Apparatus and Materials

Participants wore a Meta Quest 3 headset running in video see-through passthrough mode. Hand tracking was used to detect dealing motion and log hand kinematics. In user-controlled conditions, participants used the index trigger of a handheld controller to initiate remapping and release it to initiate recovery; the same trigger was used for the time reproduction task.

A physical table was placed in front of the participant with three marked target locations arranged in a symmetric arc (left, center, right), all comfortably within the headset field of view while requiring distinct reaching or observation directions. The distance from start to each target was 35 cm. A separate symmetric arc of three locations was marked for the experimenter, at the same distance of 35 cm. (as shown in Appx. B). Participants reported card identities after each trial via an HTML interface on a laptop (as shown in Appx. A).

We used standard playing cards restricted to number cards (2-9) in all four suits. Each condition used a fixed sequence of 9 cards, with a different sequence assigned to each of the 6 conditions. Sequences were drawn from a randomized ordering of two full decks (32 cards across 2–9 of all four suits each), yielding 54 trials per participant. Using condition-specific sequences enables consistent accuracy comparison across participants. Cards were modified to display only one suit and rank in a single corner, with the remainder occluded by white tape (Fig. 2a). As shown in prior work, playing cards provide compact, information-rich stimuli with measurable perceptual and cognitive characteristics [55], and have been used as visual stimuli across perceptual, cognitive, and psychological research [7, 35, 47, 78].

An auditory metronome at 60 bpm paced dealing so that each card was dealt in approximately one beat, keeping motion timing consistent across conditions and participants.

4.3 Time Remapping Implementation

We implemented the system on Meta Quest 3 using the Passthrough Camera API¹, which provides access to the left and right forward-facing RGB cameras via the Android Camera2 API at 1280×960 resolution and 60 Hz. Each incoming frame and its timestamp are

¹<https://developers.meta.com/horizon/documentation/spatial-sdk/spatial-sdk-pca-overview/>

stored in a circular stereo frame buffer. At each display frame, the temporal offset $d(t)$ is updated via Euler integration of the dynamical system (Eq. 4):

$$d(t + \Delta t) = d(t) + \Delta t \cdot (u(t) - \alpha(t) d(t)), \quad (6)$$

which determines the source time $g(t) = t - d(t)$ (Eq. 3). The system then retrieves the buffered frame pair whose timestamps most closely match $g(t)$ and renders it onto a head-locked canvas positioned in front of the headset, with camera extrinsics aligning the canvas to the physical scene. The remainder of the field of view displays the system passthrough. Time remapping occurs within the canvas. Since participants' viewpoints were almost stationary in both tasks, the remapped canvas was rendered without reprojecting frames to the current head pose.

Both remapping conditions used the same dynamical formulation, with the temporal offset introduced via $u(t)$ and recovered via $\alpha(t)$, varying only in how remapping was triggered. In the **user-controlled** condition, pressing the controller trigger initiated injection, and releasing it initiated recovery. In the **system-controlled** condition, remapping triggered automatically when the relevant hand departed more than 8 cm from the start location (approximately where the card's identity became visible), with recovery on return within that threshold. Parameters were fixed across all conditions: $u(t) = 0.8$ when active, and $\alpha(t)$ stepping from 0.05 to 8.0 upon recovery trigger, producing rapid resynchronization. More concretely, on trigger $u = 0.8$ sets the playback rate to $r = 1 - u = 0.2$ ($0.2 \times$ speed) and offset accumulates. The small $\alpha = 0.05$ makes catch-up weak, sustaining the slowdown. On release, u returns to 0 and α steps to 8.0, briefly fast-forwarding to clear the offset and resynchronize with the present. These were held constant to demonstrate the formalism's effectiveness without condition-specific tuning.

For the time reproduction task after each condition, a white cube appeared in front of the user for a target duration (1 s, 1.5 s, or 2 s, presented in random order across 3 trials). After the white cube disappeared, the user held the controller trigger to make a green cube appear in the same position, releasing it when they judged the same duration had elapsed. The system recorded the target duration, the reproduced duration, and the signed reproduction error for each trial.

4.4 Participants

We recruited 21 participants from the university community (10 male, 11 female; mean age = 22.48 years, $SD = 2.23$). Nineteen participants were right-hand-dominant and two were left-hand-dominant. All reported normal or corrected-to-normal vision and were able to perform the study tasks with their dominant hand. Participants received \$16 USD for approximately one hour of participation.

4.5 Procedure

Participants provided consent and completed a demographic questionnaire, then were introduced to the study structure (2 tasks $\times$ 3 conditions) via a first-person FOV recording demonstrating all three conditions and the post-condition time reproduction task. Participants were informed that time remapping would slow the visual presentation when triggered and resume toward the present

upon release. In the system-controlled condition, they were told the system would handle triggering automatically, without specifying when. After a familiarization period with the Quest 3 passthrough and task (10 min), participants performed both tasks under all three conditions (35 min), with condition order counterbalanced. In the world-centered task, the experimenter announced a verbal "3, 2, 1" countdown before each deal to signal deal onset to participants. After each condition, participants completed the time reproduction task and per-condition questionnaires. The study concluded with a semi-structured interview (15 min).

4.6 Measures

We collected objective and subjective measures after each condition. **Task accuracy** was the primary objective measure, scored at the levels of suit, rank, and full card identity. Suit and rank accuracy were scored independently per trial; full accuracy required both to be correct simultaneously. Hand kinematics were recorded throughout: in the world-centered task via the experimenter's headset, and in the body-centered task via the participant's headset. In user-controlled conditions, control activation profiles were also logged. **Motion sickness** was assessed verbally using an adapted Fast Motion Sickness scale (0–10) [39]. **Time perception** was measured via a time reproduction task (3 trials; 1 s, 1.5 s, 2 s in random order). **Subjective experience** was assessed through seven-point Likert-scale questionnaires covering discomfort, eye strain, mental demand, time pressure, frustration, perceived control, coherence, and usefulness. Finally, **semi-structured interviews** explored participants' experience and potential design dimensions and applications of time remapping.

5 Results

The results show that time remapping approximately doubled card identification accuracy relative to baseline, with gains consistent across both remapped conditions and both task types. We report kinematic analyses first to validate that dealing speed was comparable across conditions, then present accuracy results, subjective measures, and qualitative findings from post-study interviews. Each analysis uses the largest subset of valid data appropriate to the measure.

5.1 Kinematic Analysis

Experimenter Dealing Speed in the World-Centered Task.

The experimenter dealt cards at a metronome-paced rate of approximately 1.0 s per card across all observation conditions. A Friedman test confirmed no significant difference in experimenter speed across conditions ($N = 21$): $\chi^2(2, N = 21) = 2.95, p = .229, W = 0.070$. Dealing duration has $M = 1.03, 1.05, 1.05, SD = 0.14, 0.15, 0.16$ for Baseline, User-controlled, and System-controlled conditions, and all pairwise comparisons were non-significant. This confirms that any accuracy differences observed in the world-centered task cannot be attributed to variation in the stimulus presentation rate.

Participant Dealing Speed in the Body-Centered Task.

Participant dealing speed was computed as the mean per-card motion duration across trials. One participant (P20) was excluded as a speed outlier ($z = +4.78$, grand mean = 2.78 s vs. group $M = 1.34$ s, $SD = 0.30$ s; threshold: $M + 3 SD = 2.24$ s), yielding $N = 20$ for

speed analyses. P20 was subsequently excluded from the accuracy analysis.

A Friedman test across the three body-centered conditions revealed no significant overall difference in dealing speed: $\chi^2(2, N = 20) = 4.90, p = .086, W = 0.123$. Nonetheless, a consistent directional pattern emerged in post-hoc comparisons. Participants dealt approximately 250 ms slower under the system-controlled condition ($M = 1.32$ s) relative to both the baseline ($M = 1.07$ s; $\Delta = +253$ ms, $p_{\text{corr}} = .041$) and user-controlled conditions ($M = 1.14$ s; $\Delta = +183$ ms, $p_{\text{corr}} = .080$), while baseline and user-controlled speeds were virtually identical ($\Delta = +70$ ms, $p_{\text{corr}} = 1.00$).

5.2 Dynamic Acuity Performance

We evaluated card identification at three levels of granularity: full, suit, and rank accuracy (Fig. 4). P20's body-centered trials were excluded from accuracy analyses due to their classification as a speed outlier, yielding $N = 20$ for analyses of the body-center task ($N = 21$ for world-centered analyses throughout).

As accuracy distributions were non-normal and bounded, we used the Aligned Rank Transform (ART) ANOVA [76] to carry out the analysis. The result revealed a consistent main effect of condition across all measures, with no significant effect of task type and no interaction, indicating that remapping improved perceptual access uniformly regardless of whether motion was self- or externally generated. The absence of a task effect on accuracy is consistent with the kinematic finding that dealing speeds were comparable across conditions.

For **full accuracy**, there was a significant main effect of condition, $F(2, 97.02) = 116.47, p < .001$, but no effect of task type, $F(1, 98.09) = 1.34, p = .250$, and no Condition $\times$ Task interaction, $F(2, 97.02) = 0.61, p = .545$. The same pattern held for **suit accuracy**, $F(2, 97.04) = 43.42, p < .001$, and **rank accuracy**, $F(2, 97.03) = 102.09, p < .001$, with no significant main effects of task (all $p > .08$) or interactions (all $p > .25$).

Post hoc pairwise comparisons (Bonferroni-corrected) revealed that both remapped conditions significantly outperformed baseline across all measures (all $p < .001$), with no significant difference between them (all $p > .76$). Mean accuracy under remapping approximately doubled relative to baseline: full accuracy increased from $M = 0.10$ ($SD = 0.15$) at baseline to $M = 0.53$ ($SD = 0.28$) under user-controlled and $M = 0.53$ ($SD = 0.22$) under system-controlled; suit accuracy from $M = 0.32$ ($SD = 0.25$) to $M = 0.67$ ($SD = 0.20$) and $M = 0.67$ ($SD = 0.21$); and rank accuracy from $M = 0.21$ ($SD = 0.20$) to $M = 0.67$ ($SD = 0.27$) and $M = 0.73$ ($SD = 0.23$) respectively.

Notably, system-controlled remapping showed mostly lower variance than user-controlled across all measures, suggesting more uniform perceptual benefit across participants. The higher variance under user-controlled remapping likely reflects individual differences in trigger strategy and hand-eye coordination: participants who developed effective triggering strategies achieved performance superior to system-controlled, while others might struggle more with the timing demands of simultaneous dealing and trigger management.

5.3 Subjective Measures

Motion Sickness and Discomfort. We assessed motion sickness using an adapted Fast Motion Sickness (FMS) scale (0–10) [39] and additional seven-point Likert-scale ratings of discomfort and eye strain. A two-way repeated-measures ANOVA revealed a significant main effect of condition on FMS scores ($F(2, 40) = 5.57, p = .007, \eta_p^2 = 0.218$), with no effect of task and no interaction. Post-hoc comparisons showed a small increase in the system-controlled condition relative to baseline ($p = .018, d = 0.42$), while all other comparisons were non-significant. This increase may reflect the reduced predictability of trigger timing of system-controlled conditions (as discussed in Sec. 5.5).

Importantly, absolute FMS scores remained low across all conditions ($M < 1.2$), indicating negligible motion sickness in practice. Questionnaire ratings of discomfort and eye strain showed no significant effects. Together, these results suggest that temporal remapping does not induce meaningful discomfort despite altering the temporal relationship between perception and action.

Cognitive Load. We assessed cognitive load using three NASA-TLX subscales: mental demand, time pressure, and frustration (7-point scale). Repeated-measures ANOVAs revealed significant effects primarily driven by condition (as shown in Fig. 5).

Mental Demand. There were significant main effects of task ($F(1, 20) = 5.49, p = .030, \eta_p^2 = 0.215$) and condition ($F(2, 40) = 5.18, p = .010, \eta_p^2 = 0.206$), as well as a task $\times$ condition interaction ($F(2, 40) = 10.34, p < .001, \eta_p^2 = 0.341$). Overall, system-controlled remapping was rated as less mentally demanding than user-controlled ($p = .022, d = -0.451$), suggesting that offloading temporal control reduced cognitive effort. The world-centered task was also less mentally demanding than the body-centered task ($p = .030, d = -0.329$).

Time Pressure. There was a significant main effect of condition (Greenhouse-Geisser corrected: $F(1.43, 28.66) = 5.61, p = .015, \eta_p^2 = 0.219$), with Baseline rated as more time-pressured than System-controlled ($p = .049, d = 0.645$). This aligns with the accuracy results, indicating that remapping reduces perceived time pressure by extending the effective perceptual window.

Frustration. There were significant main effects of task ($F(1, 20) = 10.54, p = .004, \eta_p^2 = 0.345$) and condition ($F(2, 40) = 3.88, p = .029, \eta_p^2 = 0.163$), along with a significant task $\times$ condition interaction ($F(2, 40) = 4.16, p = .023, \eta_p^2 = 0.172$). Overall, the body-centered task was rated as more frustrating than the world-centered task ($p = .004, d = 0.501$), likely reflecting the added motor coordination demands of self-dealing.

Perceived Perceptual Benefit. Participants rated whether they could see card values more clearly than in a non-AR baseline (7-point scale) (as shown in Fig. 5). A repeated-measures ANOVA revealed a strong main effect of condition ($F(2, 40) = 45.35, p < .001, \eta_p^2 = 0.694$), with no effect of task and no interaction. Post-hoc comparisons showed that both System-controlled ($p < .001, d = 1.444$) and User-controlled ($p < .001, d = 1.587$) conditions were rated as significantly more beneficial than Baseline, with no difference between the two remapped conditions. This pattern was consistent across tasks and aligns with the objective accuracy improvements.

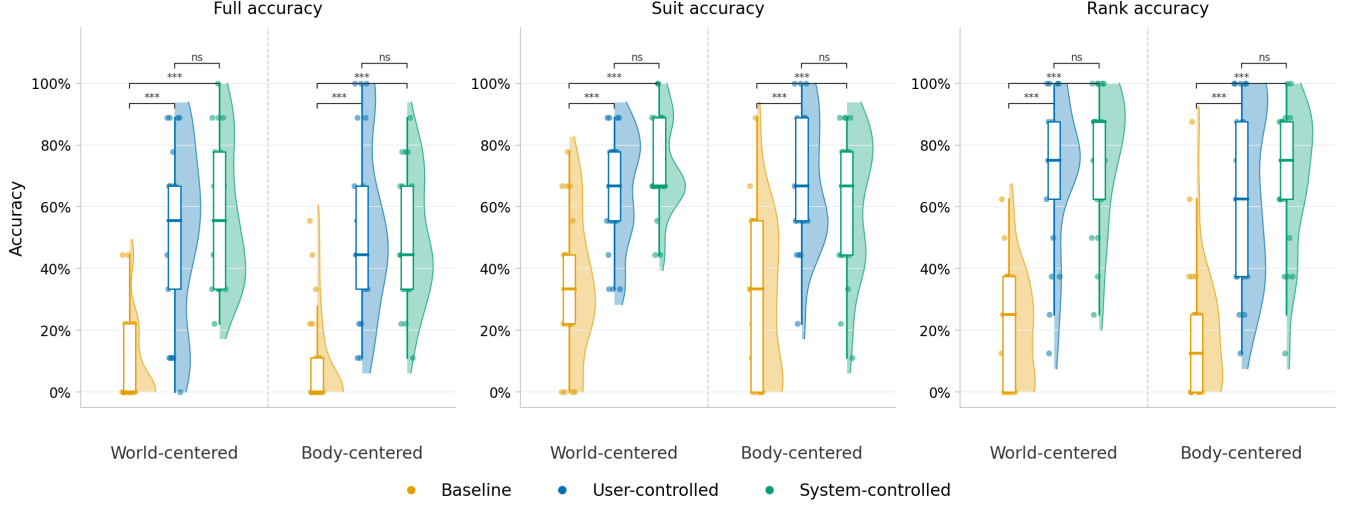


Figure 4: Graded card identification accuracy. Each panel shows the distribution of per-participant accuracy as a raincloud plot: half-violin (kernel density estimate), boxplot (median, IQR, and 1.5×IQR whiskers), and jittered individual data points ($N = 21$). Significance brackets reflect pairwise Wilcoxon signed-rank tests with Bonferroni correction ($\times 3$ per panel), consistent with the Friedman test post-hoc comparisons; $***p < .001$, $**p < .01$, $*p < .05$, ns = not significant.

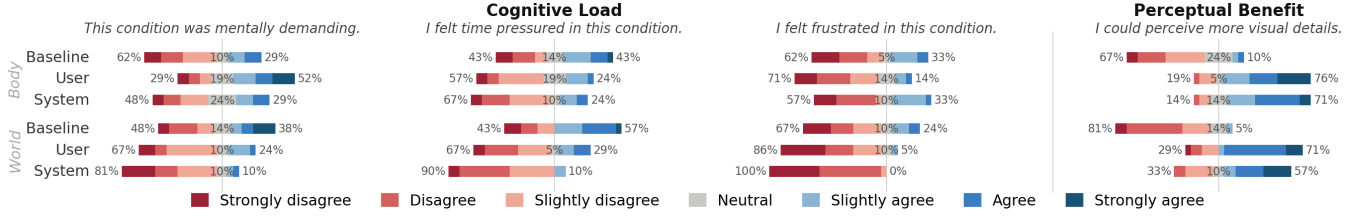


Figure 5: The Likert scale responses of all participants to 4 statements in the post-condition questionnaires.

Perceived Control and Predictability. Participants rated their sense of control over the visual experience (7-point scale). There was a significant main effect of condition ($F(2,40)=4.87$, $p=.013$, $\eta_p^2=0.196$), with User-controlled remapping rated higher than System-controlled ($p=.021$, $d=0.674$). Neither condition differed significantly from Baseline, indicating that temporal remapping does not reduce perceived control overall.

Participants also rated whether the visual experience behaved as expected. There was a main effect of task ($F(1,20)=6.22$, $p=.022$, $\eta_p^2=0.237$), with world-centered interactions rated as more predictable than body-centered ones ($p=.022$, $d=0.366$). No effects of condition or interaction were observed.

5.4 Time Reproduction Bias

Bonferroni-corrected paired t -tests revealed no significant pairwise differences. We therefore treat the following as exploratory and descriptive. The signed error showed a directional pattern: baseline body-centered reproductions were near-veridical (mean signed error = $+0.05$ s), whereas remapped conditions shifted toward underestimation (User-controlled: -0.31 s; System-controlled: -0.42 s), consistent with compression of perceived duration.

5.5 Qualitative Findings

The qualitative data were collected from semi-structured interviews. One author conducted an open thematic analysis of interview recordings, transcripts, and notes [8], with themes reviewed and refined through iterative discussions.

Temporal Anchors Extend Beyond Task Contexts. We operationalized two distinct temporal anchors through the world-centered and body-centered tasks. Participants strongly resonated with world-centered remapping for observing time-sensitive or fast events, often describing it as broadly applicable to real-world scenarios. Some also raised potential concerns, such as the possibility of exploiting slowed playback to observe sensitive actions (e.g., password entry; P10, P21). In contrast, participants emphasized that body-centered remapping fundamentally differs due to its coupling with self-generated motion. They described applications including increasing motion awareness (7/21), supporting precise motor correction (P8, P21), and creating exaggerated or dramatic effects in interactive contexts such as games (P8, P9, P20).

Beyond these task contexts, participants naturally extended the notion of temporal anchors to additional entities. Some suggested object-centered remapping, such as inducing pseudo-weight or

haptic illusions through localized temporal distortion (P15). Others proposed socially anchored remapping, where a person could serve as a temporal reference selectively remapped in others' views (P8, P13).

Control is Multi-Dimensional and Context-Dependent. Participants did not view control as a binary choice, but as a flexible continuum shaped by task demands and users' ability to anticipate events. Beyond triggering onset, some (7/21) expressed interest in controlling remapping behavior itself depending on the task context. Preferences also varied by interaction: in the world-centered task, more participants preferred system-controlled remapping (11/21 vs. 5/21), citing reliability and reduced attentional demand, whereas in the body-centered task, more preferred user control (12/21 vs. 8/21), attributing this to better anticipation and coordination with their own actions (10/21).

Participants also suggested intermediate strategies, such as pre-configured system behaviors or hybrid approaches where the system initiates remapping but allows user adjustment (P9, P10, P19, P21).

Predictability Affects User Experience. System-controlled remapping exposed participants to temporally less predictable behavior, which may have resulted in the small but significant motion sickness increase observed under system-controlled conditions (as discussed in Sec. 5.3). While some described it as uncertain or unexpected initially, they also found themselves adapting over time and becoming more comfortable across trials (7/21), suggesting that perceived transparency can evolve with experience.

Participants also noted that reduced predictability could be intentionally leveraged in certain contexts, such as motion warning or alerting systems (P7, P15, P21). But in other cases, some prefer to be well-informed of the remapping triggered or partially governed by systems (P3, P19, P21). To support understanding, they proposed additional cues to convey remapping behavior, including visual cues and overlays (P1, P19, P20, P21), auditory signals (P13), and haptic feedback (P21).

Representation Opens New Interaction Possibilities. Beyond the study implementation, participants proposed alternative representations of time remapping that enable new forms of interaction. Some suggested localized or regional effects, such as a magnifier-like view that selectively reveals remapped content (P17, P21). Others proposed decoupling the user's body from a globally remapped scene, creating a sense of "super speed" through asymmetric temporal feedback (P13, P17). Participants also envisioned multi-view representations, such as parallel windows showing alternative temporal outcomes or future predictions (P8, P10), as well as mirrored views to extend the field of view in body-centered remapping (P20).

6 Design Space

While our study directly probes anchor and control, we further distill a design space for time remapping in AR with six key dimensions based on our qualitative findings (See Fig. 6). These dimensions capture the remapping's anchor event (*Anchor*), the decision-making entity (*Control*), its implementation (*Parametrization*, *Scope*, and *Representation*), and how the effect is communicated to the user (*Transparency*). To demonstrate the feasibility of the design space,

we implemented three example applications that instantiate different combinations of the proposed dimensions.

6.1 Deriving the Design Space

Two dimensions were chosen a priori (*Anchor* and *Control*) and probed directly in our study (Sec. 3.3 and Sec. 4.1). *Parametrization* follows from the structure of the time remapping model itself (Sec. 3.1), where the parameters define how a remapping evolves over time. The remaining three dimensions were derived from the themes that emerged in our open thematic analysis of the post-study interviews (Sec. 5.5): we derived *Transparency* from participants' responses to the predictability of remapping, and *Scope* and *Representation* from their comments on where remapping applies and how it is visually presented.

6.2 Design Dimensions

Anchor. The *Anchor* defines the semantic entity whose temporal progression is remapped. This dimension is grounded in our study's task type and extended through participants' generalization of remapping targets beyond the two probed contexts. Body-centered remapping targets the user's own action: the temporal distortion is usually keyed to sensorimotor signals such as the onset and speed of the relevant body part's motion. Object-centered remapping targets specific objects or regions, enabling localized effects such as contact-dependent slowdown or pseudo-haptic impressions. World-centered remapping applies to the surrounding environment or externally generated motion, supporting observation of fast events without interfering with user action.

Control. The *Control* dimension defines who or what regulates the time remapping process. This dimension is directly probed in our study, where both objective measures and qualitative accounts revealed that control preferences may depend on the task context and users' ability to anticipate events. User-controlled remapping is explicitly initiated or adjusted by the user, providing high agency and predictability, but requiring attention and timely input. System-controlled remapping is automatically triggered based on sensed signals, events, or inferred context, reducing user effort and enabling response to fast or transient phenomena, but potentially reducing transparency or predictability. It is also well-suited for scenarios requiring synchronized experiences across users or devices. Hybrid control combines both, allowing control to be distributed between the system and the user. For example, the system can propose or initiate remapping while the user can modify or override its parameters.

Control may be realized through different input modalities (e.g., explicit input such as gestures or implicit signals such as gaze) and may operate over different control targets (e.g., onset, duration, or remapping parameters), which we treat as part of the parametrization dimension.

Parametrization. *Parametrization* defines how remapped time evolves over time. This dimension is directly grounded in the dynamical system formalization in Sec. 3.1. Key parameters include onset timing, duration, injection strength, and recovery behavior. Together, these determine the temporal profile of remapping. Different parameter configurations produce qualitatively different effects,

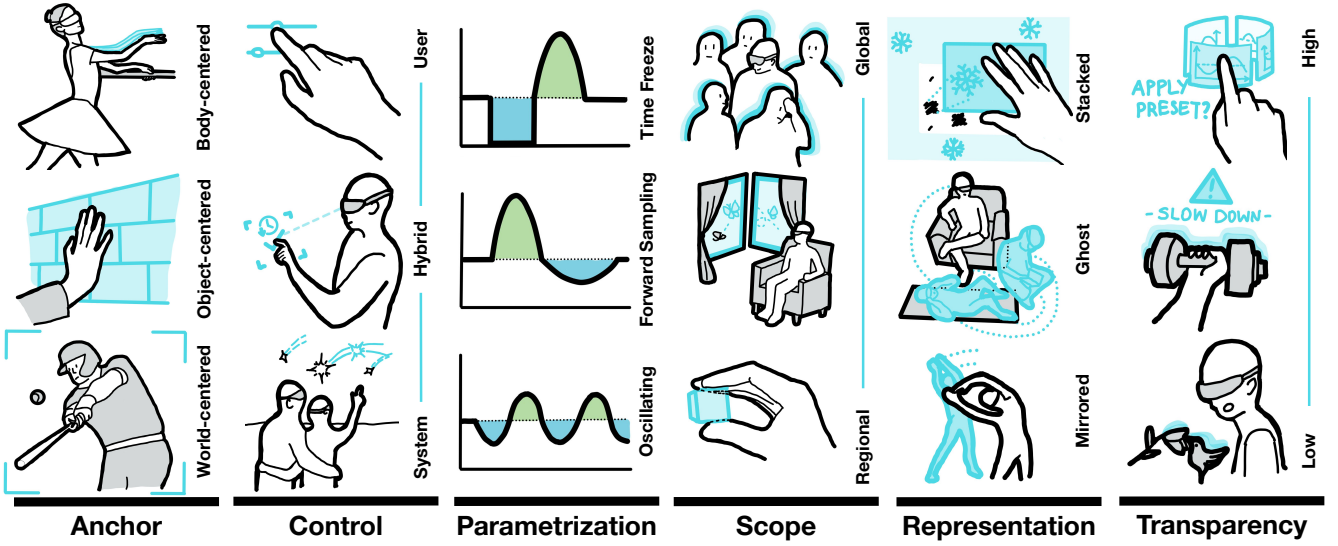


Figure 6: Among these dimensions, anchor, parametrization, and representation are discrete, each admitting a finite set of qualitatively distinct configurations, while control, scope, and transparency are better understood as continuous spectra along which designs may vary gradually.

such as brief pulses around critical events, sustained slow motion, time freezing, or adaptive profiles that vary over time.

In our study, remapping primarily manifests as slowdown with a short recovery phase. More generally, parametrization can also support acceleration (i.e., $r(t) > 1$) or mixed profiles that combine slowdown and speed-up. While speed-up can naturally exploit frames accumulated in delay, it can also be extended through forward sampling, leveraging pre-recorded or generated frames, when future motion is available in advance, particularly in repetitive motions or highly predictable scenarios.

Scope. *Scope* determines the spatial extent of temporal remapping within the visual field. This dimension emerged from participants’ proposals for localized and regional remapping effects, contrasting with the global remapping applied uniformly in our study. Remapping can be applied to localized regions, multiple regions, or the entire scene. For example, localized remapping can create fine-grained effects (e.g., pseudo-texture on a virtual object), while global remapping affects the overall temporal experience of the environment.

Representation. *Representation* determines how remapped content is visually presented. This dimension emerged from participants’ suggestions for alternative ways of presenting remapped content within the AR environment. The remapped view may fully replace the real-time view or be blended with it (e.g., as a semi-transparent ghost effect) to preserve situational awareness. Multiple representations can also be composed, such as layering different localized remappings on top of a globally remapped view.

For body-centered remapping, representation may further extend to alternative viewpoints (e.g., mirrored or third-person perspectives), potentially enabled by external sensing to overcome limitations of first-person visibility.

Transparency. *Transparency* describes how clearly the presence and behavior of time remapping are communicated to the user. This dimension emerged from participants’ varied responses to the predictability of system-controlled remapping. Under direct user control or preconfigured settings, remapping is often inherently transparent, as users can anticipate when and how it occurs. In other cases, transparency may vary from low (no explicit signaling) to high (e.g., textual, visual, auditory, or haptic cues indicating remapping state).

Transparency is also shaped by users’ mental models: as users develop an understanding of when and how the system applies remapping, the need for explicit signaling may decrease. In some scenarios, designers may intentionally reduce transparency to elicit behavioral effects (e.g., subtly influencing motion through perceptual-motor coupling), enabling applications such as speed modulation or motion guidance.

6.3 Demonstration Applications

Hybrid-Control Physical Training Speed Regulation. We implemented a speed regulation tool for physical training tasks such as weight lifting, where slower motion can improve muscle engagement. The system applies body-centered time remapping to slow down the perceived motion of the user’s actions, inducing a visual-motor mismatch that nudges the user to move more slowly and be aware of their motion. This is based on our finding that system-initiated slowdown involuntarily slowed participants’ movements (Sec. 5.1 and Sec. 7.1). A hybrid control mechanism allows the system to initiate remapping while enabling the user to override or disable it when desired.

Virtual Water with Pseudo-Texture. We implemented an object-centered remapping effect on a virtual water volume. As the user’s

hand moves through the water, temporal slowdown increases with penetration depth, creating a progressive resistance effect. Upon exiting the water, the system rapidly returns to real time. This interaction produces a pseudo-haptic illusion of viscosity or texture without physical force feedback, illustrating how localized temporal manipulation can enrich material perception.

High-Speed Slash with Forward Sampling. We implemented a high-speed slash interaction inspired by gaming scenarios. The user first performs a reference slash that the system records. During subsequent slashes, the system plays back the pre-recorded motion ahead of the user’s actual motion, creating a forward-sampled speed-up phase in which the display races ahead of real time. This is followed by a slowdown phase that allows the display to resynchronize with the user’s actual hand position. The combined effect visually exaggerates the perceived speed of the slash, amplifying the sense of impact.

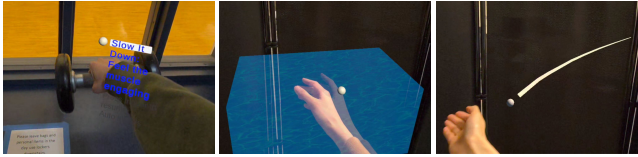


Figure 7: Screenshots from the 3 demonstration applications, physical training speed regulation tool (left), virtual water with pseudo-texture (center), and high-speed slash (right). The white dot (not visible to the user) visualizes the location of the tracked hand in real time, showing the discrepancy between the actual and displayed hand under time remapping (i.e., the displayed hand falls behind the actual hand temporally in the first and second demo, while goes ahead of the actual hand in the third). The details of the three applications are documented in the video figure.

7 Discussion

Our quantitative results show that temporal remapping can substantially improve perceptual access to fast events while remaining comfortable, and that its effects extend beyond perception into motor behavior and time perception. Our qualitative findings collectively support treating visual time as an active design material in VST AR. The following sections discuss the mechanisms and implications of these effects, the nature and limits of time remapping as a design tool, and limitations of this work.

7.1 Visuomotor Effects Under Time Remapping

Although dealing speed was controlled in our study, we observed a consistent and significant influence of **system-controlled** time remapping on participants’ motor behavior. In the body-centered task, participants dealt on average 250 ms slower under system-controlled remapping than baseline. This effect was most pronounced during practice, where almost all participants exhibited an involuntary reduction in movement speed when first encountering remapped visual feedback. This behavior suggests a coupling between the temporally altered visual presentation and motor execution. One explanation is that system-initiated remapping was

unanticipated (low transparency): lacking a prediction of the visual delay, the motor system may reflexively slow the hand to minimize the visual-proprioceptive mismatch. This might be why the effect was strongest on first exposure and diminished with practice as participants formed expectations. Although our study was not designed to isolate this mechanism, the pattern is consistent with prior work on visual dominance over proprioception in VR, including haptic retargeting [3], redirected touching [42], and pseudo-haptic feedback [45]. These observations point to time remapping as a potential tool for movement guidance, motor training, or rehabilitation, which are some applications worth direct investigation in future work.

7.2 Temporal Recalibration Under Time Remapping

In the body-centered task, remapped conditions produced a shift toward duration underestimation (User-controlled: -0.31 s; System-controlled: -0.42 s), while baseline reproductions were close to veridical ($+0.05$ s). Although these effects are not significant under Bonferroni correction, the pattern and effect sizes ($d = 0.63$ – 0.66) are consistent with prior work showing that coupling slowed visual time to self-generated action induces temporal recalibration [4]. These results should be interpreted cautiously, given that the analysis was restricted to participants with complete data ($N = 16$). While time perception was an exploratory measure in our study, future work can investigate further how body-centered time remapping may help study and design temporal recalibration effects in AR.

7.3 The Nature and Limits of Time Remapping

Time remapping operates on the visual channel alone, while other modalities (e.g., auditory, haptics) remain anchored to physical time. Its effectiveness therefore depends on visual dominance over competing signals. This dominance is well-supported in static or low-demand contexts, but may weaken under conditions of high proprioceptive load, fast reactive motion, or strong auditory temporal cues. Designers should therefore consider the context of time remapping carefully, adjusting control level, scope, and parametrization to maintain cross-modal coherence when necessary. Extending time remapping to additional modalities or designing representations that reduce cross-modal salience are also promising directions for future work.

Beyond effectiveness, designers should also consider when to withhold remapping. If such capability becomes widely available, users may develop inflated expectations of perceptual access to fast events, similar to how constant photo availability has been argued to reduce investment in unmediated observation [34]. Preserving the experience of genuinely fleeting moments is as important a design consideration as extending them.

7.4 Safety Considerations Under Remapped Time

Time remapping decouples the display from the present, so it might pose a risk to users who rely on the display to act. A past offset ($d > 0$) lags reality while a forward-sampled future ($d < 0$) leads it, so the user may act on an event that has not yet happened,

or that the prediction gets wrong. In our high-speed slash demo, for instance, the user does not see their real hand throughout the slash, which may reduce awareness of their body and surroundings and should therefore be performed with caution. More generally, designs that withhold real-time information, such as body position or the surroundings for a prolonged period require careful consideration: they should be confined to predictable motions and unobstructed workspaces, use conservative prediction horizons, or provide accessible complementary real-time cues.

7.5 Limitations and Future Work

Playing Cards as Visual Stimuli. On playing cards, the perceptual difficulty was not uniformly distributed across identities (e.g., clubs and spades are harder to differentiate than hearts and diamonds). In addition, because symbols were placed at the top-left corner, the task introduced an asymmetry for left-handed participants (2/21), who experienced increased difficulty due to occlusion and dealing direction.

Speed Control and Individual Differences. Although no significant condition effects on dealing speed were observed, substantial individual variability remained. While a metronome provided pacing guidance, differences in motor execution and coordination led to variability in effective stimulus exposure. We consider this as an inherent trade-off when studying perception in action-coupled settings.

Hardware and Software Limitations. Meta’s Passthrough Camera API limits resolution (up to 1280×1280), frame rate (60 Hz), and precise control over camera-to-display mapping. Additionally, the API does not support regional or selective camera texture modification, so the body-centered conditions and object- and body-centered demo implementations apply time remapping to the full canvas instead of a localized region. These limitations may affect visual fidelity and the perceived effectiveness of time remapping.

Counterbalancing. In our study, conditions were balanced as evenly as possible among all participants, but not perfectly counterbalanced, so residual order effects within a task cannot be fully excluded.

Long-Term Effects. Our evaluation focused on short-term use (9 trials per condition per task), leaving questions of long-term adaptation, fatigue, and everyday usability open. While participants reported that extended practice might improve performance and refine strategies, such effects were not directly measured. Some participants suggested that time remapping may have potential benefits for memory enhancement and accessibility. Future work may validate and examine additional perceptual and cognitive effects over longer-term use.

Motion-based Time Reproduction. We measured time perception with a standard static duration-reproduction task from psychophysics. However, since our effects are motion-coupled, time reproduction tasks based on moving or action-coupled stimuli could be a natural, likely more sensitive next step.

Head Reprojection. Time remapping controls which captured moment of the scene is displayed, which is independent of reprojection the displayed frame to the user’s current head pose to compensate for head motions. Because remapping can display past

or future frames, a moving head would see spatial misregistration, which could potentially lead to vestibular-visual mismatch. Positional/asynchronous time warp [71] is a technique on commodity headsets² that can address this issue. In our study, the tasks were near-stationary, so we rendered a head-locked canvas without reprojection (Sec. 4.3). However, more dynamic scenarios might require composing remapping with head reprojection to preserve registration and comfort.

8 Conclusion

This paper introduced time remapping as a design material for VST AR, formalized as a first-order dynamical system. A controlled within-subjects study ($N = 21$) showed that time remapping approximately doubles dynamic acuity without meaningfully increasing discomfort, with gains consistent across user- and system-controlled conditions and both task types. Beyond perceptual performance, body-centered remapping produced visuomotor coupling effects and a directional trend toward duration underestimation, pointing to motor and temporal recalibration as secondary effects worth direct investigation. Building on qualitative findings, we derived a six-dimensional design space (*Anchor, Control, Parametrization, Scope, Representation, and Transparency*) instantiated through three demonstration applications. More broadly, this work positions programmable time as a new design dimension for VST AR and suggests that understanding when and how to extend, compress, or preserve visual time will be essential to designing temporally meaningful AR experiences.

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²<https://developers.meta.com/horizon/blog/asynchronous-timewarp-examined/>

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A HTML Interface for Card Identification Report

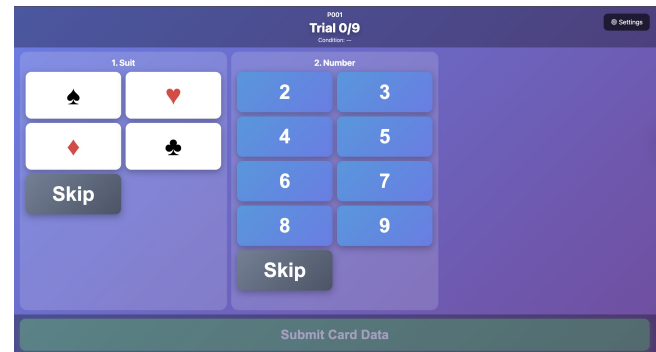


Figure 8: Participants make selections of which suit and rank they see in a trial on the interface through a Macbook Air. If they choose “Skip” for suits, it will prompt them to select the color they saw. They may also skip the selection of color. The icons are highly discernible in Meta Quest 3 passthrough.

B Detailed Layout of the Study

All start and target pairs are of a 35 cm distance. The experimenter’s target spread a smaller arc due to the fact that they are closer to the participants’ headset and need to fit within the remapped canvas. A tripod holds the card deck for the participants in all conditions, as they need to use one hand to deal and the other to control in user-controlled conditions.

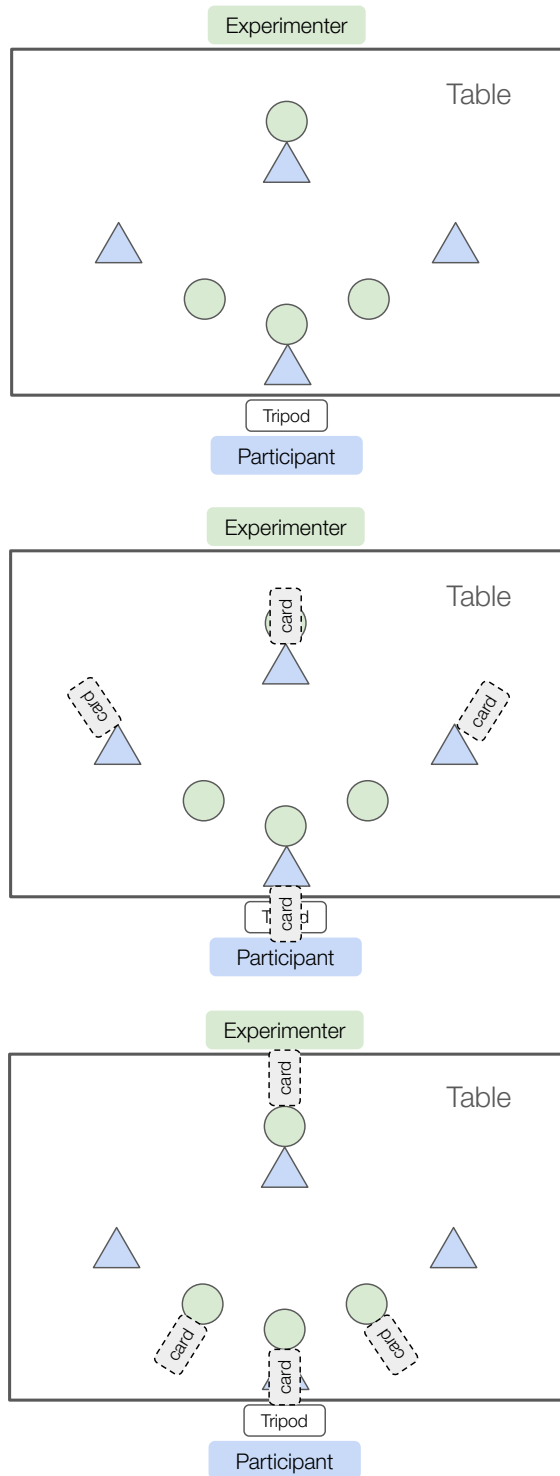


Figure 9: An illustration of the exact study setup (top). The green circles correspond to the experimenter's start and target locations, while the blue triangles are the participants' start and target marked locations. The exact card positions (start and end) in the body-centered (middle) and world-centered (bottom) tasks are also illustrated. Card suits and ranks can be seen clearly when static even placed at the furthest possible point (the experimenter's start point).